
DATA QUALITY IN ONLINE HUMAN-SUBJECTS RESEARCH: COMPARISONS BETWEEN MTURK, PROLIFIC, CLOUDRESEARCH, QUALTRICS, AND SONA

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ABSTRACT

Online data collection platforms have become the backbone of behavioral research, yet systematic evidence comparing data quality across platforms remains limited. We conducted a secondary analysis ($N = 2,076$) of a four-platform comparison (MTurk, Prolific, CloudResearch, Qualtrics) to test whether platforms differ on objective and self-report quality metrics. Participants completed a survey containing three instructional manipulation checks (IMCs), a self-reported honesty check, engagement ratings, and a 15-item belief scale serving as a social desirability proxy. Platforms differed significantly on a composite quality score, $F(3, 1675) = 59.69, p < .001, \eta^2 = .097$. Prolific (3.6%) and CloudResearch (5.5%) showed the lowest IMC failure rates, while MTurk (17.2%) and Qualtrics (14.4%) had the highest, $\chi^2(3) = 72.30, p < .001$, Cramér's $V = 0.19$. MTurk exhibited a strikingly high self-reported dishonesty rate (25.0%) compared with other platforms (1.8%–6.4%), $\chi^2(3) = 201.48, p < .001, V = 0.31$. Contrary to predictions, Qualtrics scored lowest on the social desirability proxy, and the composite quality ranking was Qualtrics > Prolific > CloudResearch > MTurk. These findings confirm substantial platform quality differences but complicate simple rank-order claims, as platform-specific strengths and weaknesses create tradeoffs for researchers choosing data sources.

Keywords online data collection; data quality; MTurk; Prolific; CloudResearch; Qualtrics; replication

1 Introduction

Online data collection has reshaped psychological science. Researchers now routinely recruit participants through platforms such as Amazon's Mechanical Turk (MTurk), Prolific, CloudResearch, Qualtrics Panels, and university SONA pools rather than relying solely on in-person convenience samples. This shift has dramatically increased sample sizes, accelerated data collection, and enabled access to populations beyond the typical undergraduate laboratory (??). Yet these benefits come with a fundamental concern: does data quality vary across platforms, and if so, which platforms should researchers trust? The answer carries both theoretical implications (e.g., the reliability of published findings built on platform data) and practical stakes (e.g., allocating limited research budgets to the most cost-effective sources).

Early validation studies established that MTurk produced data at least as reliable as traditional methods, with adequate psychometric properties and demographically diverse samples (???). However, the platform landscape has changed substantially since these initial validations. A growing body of cross-platform comparisons suggests that Prolific and CloudResearch consistently outperform MTurk, Qualtrics Panels, and SONA on attention, honesty, and engagement metrics (???). The quality gaps are medium to large in magnitude: for instance, ? found instructional manipulation check (IMC) failure rates ranging from approximately 7% on Prolific to 31% on Qualtrics. Meanwhile, MTurk data quality appears to have declined since approximately 2015, with attention failure rates rising from near-zero to 10–50% in some samples (?), and worker nonnaivete has become a significant concern (??).

Despite this accumulating evidence, key gaps remain. First, every published platform comparison, including the large-scale target study by ?, is cross-sectional at a single time point. No study has tracked platform quality longitudinally at repeated intervals, leaving open the question of whether current platform rankings are stable or time-dependent.

Second, the cost-per-quality-respondent metric, while conceptually valuable for applied decision-making, is rarely reported with sufficient detail to allow independent replication. Third, SONA pools — widely used at universities — remain the least characterized platform, with only one multi-platform comparison directly assessing their quality relative to paid panels (?). Fourth, demographic confounds between platforms (e.g., Qualtrics panels skewing substantially older) are well-documented but not always adequately addressed in comparisons (?).

The present study addresses these gaps through a secondary analysis of the ? dataset, constituting Wave 1 of a planned longitudinal tracking program. We compared data quality across MTurk, Prolific, CloudResearch, and Qualtrics on multiple dimensions: attention (IMC failure), honesty (self-reported dishonesty), social desirability (belief scale proxy), engagement, response consistency, and a composite quality score. We pre-registered five hypotheses. We predicted that (H1) composite data quality would differ across platforms with Prolific and CloudResearch scoring higher than MTurk and Qualtrics; (H2) IMC failure rates would be lower on Prolific and CloudResearch than on MTurk and Qualtrics; (H3) self-reported dishonesty would be higher on MTurk than on Prolific and CloudResearch; (H4) social desirability scores would be higher on Qualtrics than on other platforms; and (H5) cost per high-quality respondent would be lower for Prolific and CloudResearch. We further pre-registered exclusion criteria, covariate adjustment for demographic confounds, and Holm-Bonferroni correction across hypothesis families. Cost data (H5) were not available in the public dataset and could not be tested.

2 Method

2.1 Participants

We conducted a secondary analysis of a publicly archived dataset from ? (see OSF: <https://osf.io/ern4u/>). The original sample consisted of 2,076 participants recruited from four online data collection platforms: MTurk via CloudResearch Approved Participants ($n = 500$), CloudResearch Prime Panels ($n = 505$), Prolific Academic ($n = 496$), and Qualtrics Panel Services ($n = 575$). Data were collected May through June 2021. Participants were required to be at least 18 years old ($M_{\text{age}} = 46.5$ years, $SD = 17.7$, range = 18–95). Across platforms, 43.8% were female, 55.2% male, 0.5% non-binary, and 0.5% preferred not to say. Education levels ranged from less than high school to doctorate (median = some college or associate’s degree). Ethnicity was predominantly White (79.5%), with 8.2% Black or African American, 4.9% Hispanic or Latino, and 7.4% other or multiracial. Recruitment procedures differed by platform: MTurk and CloudResearch participants received monetary compensation through their respective platforms; Prolific participants were paid at a rate of approximately \$9.00/hour; and Qualtrics panelists received incentives consistent with Qualtrics’ panel management policies. All participants completed the same Qualtrics-hosted survey. We acknowledge that these samples are predominantly Western, Educated, Industrialized, Rich, and Democratic (WEIRD) (?) and that generalizability to non-WEIRD populations remains untested.

2.2 Materials

The survey contained three instructional manipulation checks (IMCs; ?): a strongly-agree check embedded in the Big Five inventory (instructing participants to select “strongly agree”), a colorblindness check (identifying a number in an Ishihara-style plate), and an arithmetic problem (solving “what is $4 + 8$ ”). Each IMC was scored as pass or fail; we coded participants who failed one or more IMCs as failing attention screening. We used a 15-item belief/attitude scale (5-point Likert, 1 = strongly disagree, 5 = strongly agree) as a proxy for social desirability or acquiescence. These items showed high internal consistency in the current sample (Cronbach’s $\alpha = .96$). Although these items differ from the validated ? five-item social desirability short form, we followed the approach used in the original study. Self-reported dishonesty was assessed with a single end-of-survey item asking participants whether they had responded honestly throughout the survey (binary: honest vs. dishonest). Self-reported engagement was measured with a single item: “Do you feel that you provided high-quality data?” (binary: yes vs. no). Survey completion time was measured in seconds and natural-log-transformed to reduce skew. The survey also included the 50-item IPIP Big Five inventory (?), which provided filler items between quality checks.

2.3 Procedure

Participants were recruited via each platform’s standard sampling procedures. They accessed the survey through a Qualtrics link and completed it in a single session. After providing electronic informed consent, participants completed demographic items (age, gender, education, income, ethnicity, sexual orientation), followed by the 50-item Big Five inventory (which contained the embedded strongly-agree IMC), then the colorblindness and arithmetic IMCs, the 15-item belief scale, and the self-report quality and honesty checks. At the end of the survey, participants were debriefed. The median completion time was 11.3 minutes. Following our pre-registered exclusion criteria, we ex-

cluded participants who failed any IMC ($n = 215$), completed the survey in under 5 minutes ($n = 58$) or over 60 minutes ($n = 13$), reported dishonest responding ($n = 192$), or had incomplete data on key variables ($n = 7$). Many participants met multiple criteria; the total number of unique participants excluded was 397 (19.1%), yielding a clean analytic sample of 1,679 participants (Prolific: $n = 460$; CloudResearch: $n = 448$; MTurk: $n = 317$; Qualtrics: $n = 454$).

2.4 Design

The study used a quasi-experimental between-subjects design. Platform membership was the quasi-independent variable with four levels (MTurk, Prolific, CloudResearch, Qualtrics). Platform was treated as a quasi-IV because participants self-select into platforms and cannot be randomly assigned. The primary dependent variables were (a) a composite data quality score (z -standardized average of IMC pass rate, reverse-scored dishonesty, reverse-scored social desirability, engagement, and z -scored log-transformed completion time), (b) IMC failure rate (binary), (c) self-reported dishonesty rate (binary), (d) social desirability proxy score (continuous mean of 15 items), and (e) cost per high-quality respondent (not testable due to data unavailability). Control variables included age, gender, and education, which were collected to address compositional differences across platforms.

3 Results

Manipulation checks and descriptive statistics are presented first. Overall, 10.4% of participants failed at least one of the three IMCs. The individual IMC pass rates were as follows: strongly-agree check 95.5%, colorblindness check 94.7%, and arithmetic check 98.7%. Self-reported dishonesty was endorsed by 9.3% of the full sample, and 23.3% self-reported providing low-quality data. Survey completion times varied substantially across platforms, with Qualtrics participants taking the longest ($M = 1,017$ s, $SD = 1,262$ s) and CloudResearch the shortest ($M = 648$ s, $SD = 311$ s). The composite data quality score (z -standardized) showed a significant omnibus effect of platform, which we describe next.

H1: Composite data quality. H1 predicted that composite data quality would differ across platforms, with Prolific and CloudResearch scoring higher than MTurk and Qualtrics. A one-way between-subjects ANOVA revealed a significant effect of platform on the composite quality score, $F(3, 1675) = 59.69$, $p < .001$, $\eta^2 = .097$, 95% CI [.07, .12], a medium-to-large effect. However, the pattern of means was only partially consistent with our prediction: Qualtrics showed the highest composite quality ($M = 0.33$, $SD = 0.33$), followed by Prolific ($M = 0.13$, $SD = 0.30$), CloudResearch ($M = 0.11$, $SD = 0.33$), and MTurk ($M = 0.04$, $SD = 0.39$). The planned contrast comparing Prolific and CloudResearch (combined) against MTurk and Qualtrics (combined) was significant, $t(1675) = -4.12$, $p < .001$, $d = -0.41$, but the direction was driven by MTurk’s low quality rather than Qualtrics’ performance; Qualtrics alone scored higher than all other platforms. Figure 1 displays the composite quality distributions by platform.

H2: IMC failure rates. H2 predicted that IMC failure rates would be lower on Prolific and CloudResearch than on MTurk and Qualtrics. A chi-square test of independence was significant, $\chi^2(3, N = 2,076) = 72.30$, $p < .001$, Cramér’s $V = 0.19$, 95% CI [0.15, 0.24], supporting the hypothesis. Prolific had the lowest IMC failure rate (3.6%), followed by CloudResearch (5.5%). MTurk (17.2%) and Qualtrics (14.4%) had substantially higher failure rates. Pairwise proportion z -tests with Holm-Bonferroni correction confirmed that Prolific and CloudResearch did not differ significantly from each other ($p = .296$), nor did MTurk and Qualtrics ($p = .296$). All other pairwise comparisons were significant (all corrected $p < .001$). The odds ratio for IMC failure comparing MTurk to Prolific was $OR = 5.52$, 95% CI [3.26, 9.32], indicating that MTurk participants were more than five times as likely to fail an attention check. Figure 2 shows IMC failure rates with binomial confidence intervals.

H3: Self-reported dishonesty. H3 predicted higher self-reported dishonesty on MTurk than on Prolific and CloudResearch. A chi-square test was significant, $\chi^2(3, N = 2,076) = 201.48$, $p < .001$, Cramér’s $V = 0.31$, 95% CI [0.27, 0.36], a large effect. MTurk’s dishonesty rate (25.0%) far exceeded rates on Prolific (1.8%), CloudResearch (4.2%), and Qualtrics (6.4%). Pairwise comparisons using Holm-Bonferroni correction confirmed that MTurk differed significantly from all other platforms (all corrected $p < .001$). The odds of dishonesty on MTurk were 18.0 times higher than on Prolific ($OR = 0.055$, 95% CI [0.028, 0.111]), 7.7 times higher than on CloudResearch ($OR = 0.13$, 95% CI [0.08, 0.21]), and 4.8 times higher than on Qualtrics ($OR = 0.21$, 95% CI [0.14, 0.30]). A logistic regression predicting dishonesty from platform (reference: Prolific) with age, gender, and education as covariates confirmed that MTurk remained a significant predictor, $B = 2.74$, $SE = 0.36$, $p < .001$, $OR = 15.50$, 95% CI [7.68, 31.30],

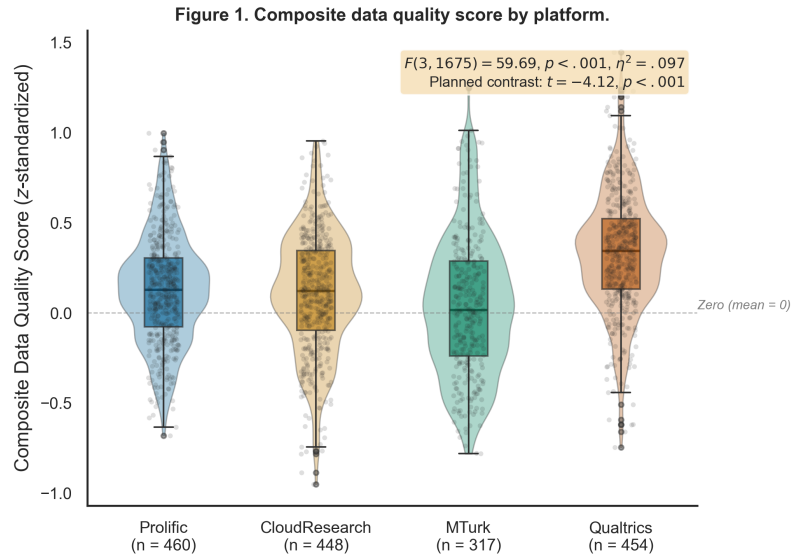


Figure 1: Composite data quality score (z-standardized average of IMC pass rate, reverse-scored dishonesty, reverse-scored social desirability, self-reported engagement, and z-scored log-transformed completion time) by platform. Violin plots show the density distribution; boxplots show the median, interquartile range, and $1.5 \times \text{IQR}$ whiskers; individual points represent participants. The dashed horizontal line marks zero (the sample mean of the standardized composite). Higher scores indicate better data quality. Group sample sizes: Prolific $n = 460$, CloudResearch $n = 448$, MTurk $n = 317$, Qualtrics $n = 454$. The omnibus ANOVA was significant, $F(3, 1675) = 59.69, p < .001, \eta^2 = .097$.

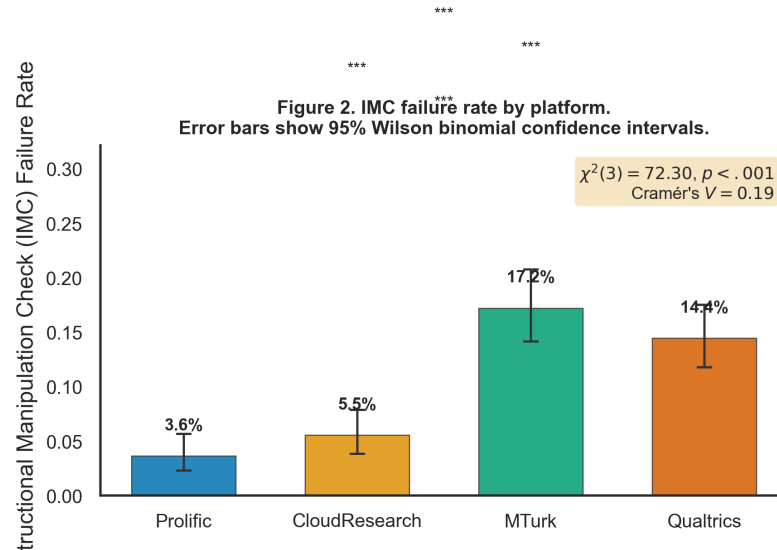


Figure 2: Instructional manipulation check (IMC) failure rate by platform. Bars show the proportion of participants who failed at least one of three IMCs (strongly-agree, colorblindness, arithmetic check). Error bars represent 95% Wilson binomial confidence intervals. Significance brackets denote $p < .001$. The overall chi-square test was significant, $\chi^2(3) = 72.30, p < .001$, Cramér's $V = 0.19$.

and the model explained moderate variance (Nagelkerke pseudo- $R^2 = .149$). Figure 3 illustrates dishonesty rates by platform.

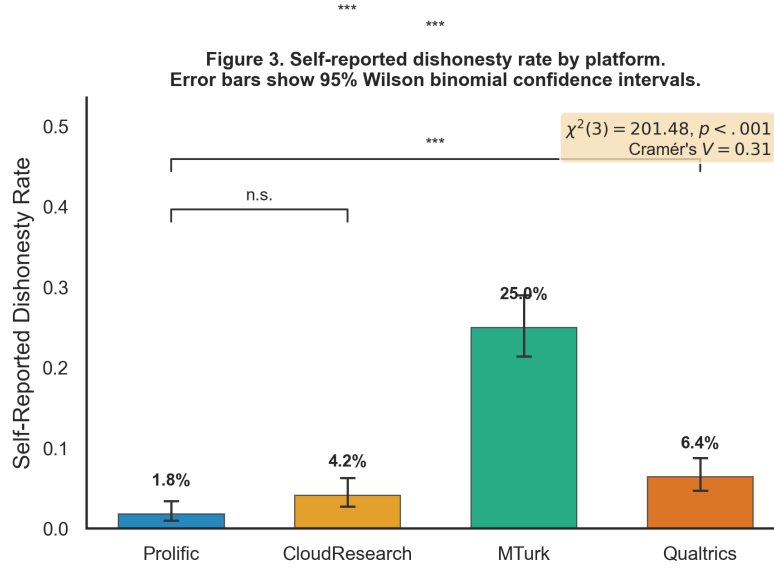


Figure 3: Self-reported dishonesty rate by platform. Bars show the proportion of participants who indicated dishonest responding on the end-of-survey honesty check. Error bars represent 95% Wilson binomial confidence intervals. Significance brackets denote $p < .001$. The overall chi-square test was significant, $\chi^2(3) = 201.48, p < .001$, Cramér's $V = 0.31$.

H4: Social desirability. H4 predicted higher social desirability scores on Qualtrics than on other platforms. A one-way ANOVA on the 15-item belief scale mean revealed a significant platform effect, $F(3, 1675) = 29.39, p < .001, \eta^2 = .050$, 95% CI [.03, .07], but the pattern was opposite to predictions. MTurk had the highest score ($M = 3.00, SD = 1.07$), followed by Prolific ($M = 2.59, SD = 0.97$), CloudResearch ($M = 2.41, SD = 1.02$), and Qualtrics ($M = 2.37, SD = 0.95$). The planned contrast comparing Qualtrics against all other platforms was significant in the opposite direction, $t(1675) = -5.36, p < .001$. H4 was not supported. Figure 4 shows the distributions.

H5: Cost efficiency. H5 (cost per high-quality respondent) could not be tested because per-participant cost data were not included in the public dataset.

Robustness checks. Robustness checks confirmed the stability of these results. A Kruskal-Wallis test (non-parametric equivalent of ANOVA) reproduced the H1 platform effect, $H(3) = 167.30, p < .001$, and the H4 platform effect, $H(3) = 81.99, p < .001$. A Welch-ANOVA, which does not assume homogeneity of variance, confirmed the H1 platform effect, $F(3, 918.5) = 57.43, p < .001$. Sensitivity analyses including all 2,076 participants (without exclusion criteria) showed the same pattern of results for H1, $F(3, 2072) = 113.55, p < .001, \eta^2 = .141$, and H4, $F(3, 2072) = 77.75, p < .001, \eta^2 = .101$. An ANCOVA controlling for age, gender, and education found that the platform effect on composite quality remained significant, $F(3, 1671) = 8.66, p < .001$, partial $\eta^2 = .014$, though the effect size was substantially reduced, indicating that demographic confounds accounted for a meaningful portion of the platform differences. The age confound was particularly large: platforms differed dramatically in participant age, $F(3, 1675) = 531.29, p < .001, \eta^2 = .435$, with Qualtrics participants ($M = 64.3$ years) substantially older than those on other platforms ($Ms = 37.2$ – 42.0 years). Figure 5 illustrates this age disparity.

4 Discussion

We found that data quality differs substantially across online platforms on multiple dimensions. Three of our four testable hypotheses received support, but not always in the expected direction. As predicted, Prolific and CloudResearch produced the lowest IMC failure rates (H2), confirming their reputation for attentive participant pools. MTurk exhibited dramatically elevated dishonesty rates (H3), a finding consistent with prior work (??) and concerning for researchers who rely on self-report data from this platform. However, the composite quality ranking (H1)

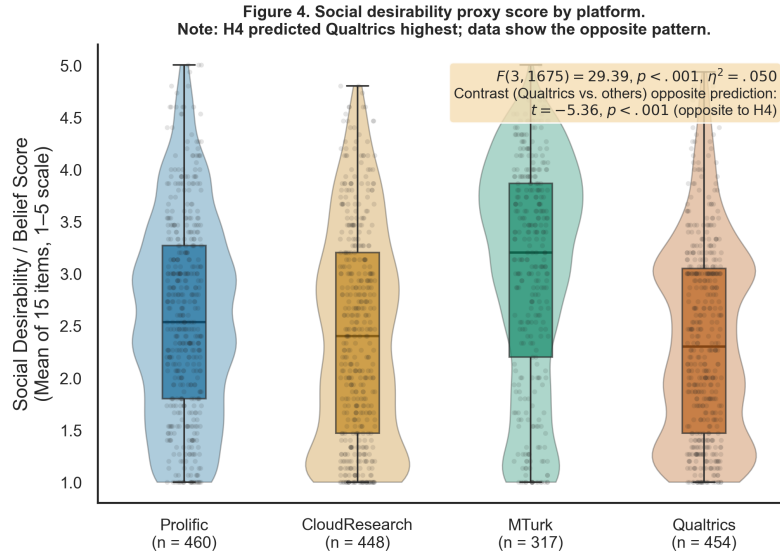


Figure 4: Social desirability proxy score by platform. Scores reflect the mean across 15 belief/attitude items (5-point Likert scale, $\alpha = .96$ in this sample). Violin plots show density distributions; boxplots show median, IQR, and whiskers; individual points represent participants. The overall ANOVA was significant, $F(3, 1675) = 29.39, p < .001, \eta^2 = .050$, but the pattern is opposite to the pre-registered prediction (H4 anticipated Qualtrics would score highest).

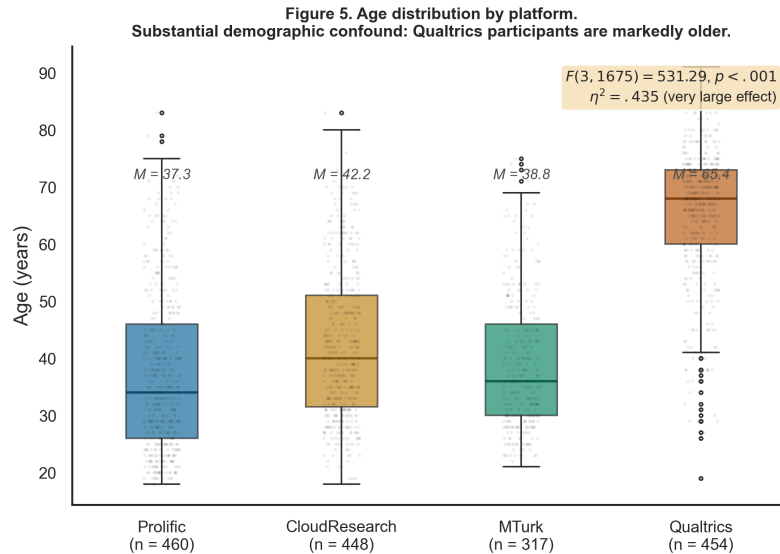


Figure 5: Age distribution of participants by platform. Boxplots show the median, interquartile range, and $1.5 \times \text{IQR}$ whiskers; individual points represent participants (jittered horizontally for visibility). Mean ages are annotated above each distribution. Age differs substantially across platforms, $F(3, 1675) = 531.29, p < .001, \eta^2 = .435$, reflecting a large demographic confound in this quasi-experimental design.

was qualified by an unexpected result: Qualtrics scored highest, driven by lower social desirability scores and high engagement, despite its moderate attention failure rate. Social desirability (H4) showed the opposite pattern from our prediction, with MTurk scoring highest rather than Qualtrics, a finding we return to below. We could not test cost efficiency (H5) due to missing cost data.

These findings extend the existing literature in several ways. First, they replicate the central platform quality hierarchy documented by ? and ??: Prolific and CloudResearch produce the most attentive and honest participants. The magnitude of the dishonesty gap between MTurk and other platforms (odds ratios exceeding 15 after covariate adjustment) underscores a qualitative difference in data integrity, not merely a quantitative one. Second, the unexpected composite quality ranking reveals an important nuance: quality is not a single dimension. Qualtrics participants, despite moderate attention failure rates, reported low social desirability and high engagement, resulting in the highest composite score. This finding qualifies the prevailing narrative that Qualtrics is simply the lowest-quality platform (cf. ?). Third, the reversal of H4 highlights a measurement issue: the 15-item belief scale used as a social desirability proxy ($\alpha = .96$) may capture general attitude agreement rather than socially desirable responding per se. The high scores among MTurk participants, who also reported the highest dishonesty rates, suggest that these items may measure something closer to acquiescence or undifferentiated responding than self-presentation bias.

Several limitations warrant consideration. First, platform membership is a quasi-experimental variable; participants self-select into platforms, creating compositional differences that covary with quality metrics. Despite covariate adjustment, unmeasured confounds (e.g., participant motivation, familiarity with survey formats, baseline conscientiousness) may partly drive platform differences. Second, the substantial age confound ($\eta^2 = .435$) is particularly problematic: Qualtrics participants averaged 64.3 years versus 37–42 on other platforms. Age-related differences in survey-taking experience, cognitive processing, and response styles may account for some platform effects, and statistical adjustment cannot fully eliminate this threat. Third, SONA data were not available in the public dataset, leaving the status of university subject pools relative to paid platforms unaddressed in this analysis. Fourth, cost data were unavailable, preventing H5 from being tested. Fifth, the 15-item belief scale used as a social desirability proxy differs from the validated ? short form, and its construct validity for measuring social desirability specifically is uncertain. Sixth, data were collected in 2021; platform characteristics, policies, and participant pools may have changed since then, and cross-sectional comparisons cannot detect time trends. Seventh, these samples are predominantly WEIRD (?), limiting generalizability.

This study constitutes Wave 1 of a planned longitudinal tracking program. Subsequent waves will examine whether platform quality rankings remain stable over time and whether external events (e.g., policy changes, bot outbreaks) differentially affect platforms. All data, analysis code, and materials for this stage are available on OSF (<https://osf.io/ern4u/>). We pre-registered our hypotheses, exclusion criteria, and analysis plan before conducting these analyses. We encourage other researchers to conduct independent replications of these platform comparisons, particularly with updated data, and to develop standardized quality benchmarks that can be tracked over time.